

Equation of motion

$$h[k+1] = h[k] + a_h[k]\{f_{dh}[k] + f_T[k]\} + \delta h_D[k],$$

Motion error

$$\delta h[k+1] = \delta h[k] + \Delta h_d[k] - a_h[k]\{f_{dh}[k] + f_T[k]\} - \delta h_D[k],$$

$\delta h[k] = h_d[k] - h[k]$, $h_d[k]$ is the desired motion trajectory, $\delta h_D[k]$ is the motion error caused by the disturbance, and $\Delta h_d[k] = h_d[k+1] - h_d[k]$ is the desired motion step size.

Measurement

(d -step delay)

$$\delta h_m[k] = \delta h[k-d] + n_h[k],$$

Control objective: $\delta h[k+1] = \lambda_c \cdot \delta h[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dh}[k] = a_h^{-1}[k]\{\Delta h_d[k] + (1 - \lambda_c)\delta h[k] - \delta h_D[k]\}$$

Motion error:

$$\delta h[k+1] = \delta h[k] + \Delta h_d[k] - a_h[k]\{\bar{f}_{dh}[k] + f_T[k]\} - \delta h_D[k] = \lambda_c \cdot \delta h[k] - a_h[k]f_T[k]$$

d -step measurement delay:

$$\delta h[k] = \delta h[k-d] + \sum_{i=1}^d \Delta h_d[k-i] - \sum_{i=1}^d a_h[k-i]\{f_{dh}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta h_D[k-i]$$

Ideal control effort (with d -step measurement delay):

$$\begin{aligned} \bar{\bar{f}}_{dh}[k] = a_h^{-1}[k] \Big\{ & \underbrace{\left[\Delta h_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta h_d[k-i] \right]}_{\Delta h_d^d[k]} + (1 - \lambda_c) \left[\delta h[k-d] - \sum_{i=1}^d a_h[k-i] f_{dh}[k-i] \right] \\ & - \underbrace{\left[\delta h_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta h_D[k-i] \right]}_{\delta h_D^d[k]} \Big\} \end{aligned}$$

Motion error:

$$\delta h[k+1] = \lambda_c \cdot \delta h[k] - \underbrace{\left\{ a_h[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_h[k-i]f_T[k-i] \right\}}_{\delta h_T^d[k]}$$

Implementable control effort

$(\delta \hat{h}_D^d \equiv 0$: disturbance not estimated)

$$f_{dh}[k] = \hat{a}_h^{-1}[k] \left\{ \Delta h_d^d[k] + (1 - \lambda_c) \left[\delta h_m[k] - \sum_{i=1}^d \hat{a}_h[k-i] f_{dh}[k-i] \right] \right\}$$

Motion error

$$\delta h[k+1] = \lambda_c \cdot \delta h[k] - a_h[k] [f_{dh}[k] - \bar{\bar{f}}_{dh}[k]] - \delta h_T^d[k]$$

$$a_h[k] \bar{\bar{f}}_{dh}[k] = \Delta h_d^d[k] + (1 - \lambda_c) \left[\delta h[k-d] - \sum_{i=1}^d a_h[k-i] f_{dh}[k-i] \right] - \delta h_D^d[k]$$

$$a_h[k] f_{dh}[k] = (1 + e_{a_h}[k] \hat{a}_h^{-1}[k]) \left\{ \Delta h_d^d[k] + (1 - \lambda_c) \left[(\delta h[k-d] + n_h[k]) - \sum_{i=1}^d \hat{a}_h[k-i] f_{dh}[k-i] \right] \right\}$$

$$\begin{aligned} a_h[k] f_{dh}[k] - a_h[k] \bar{\bar{f}}_{dh}[k] &= e_{a_h}[k] f_{dh}[k] + (1 - \lambda_c) n_h[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_h[k-i] - \hat{a}_h[k-i]) f_{dh}[k-i] + \delta h_D^d[k] \end{aligned}$$

Gain model (exact curve; definitions only):

$$\begin{aligned} a_h[k] &= a_h(h_d[k]) - a'_{hd}[k] \delta h[k], & a_h(h_d[k+1]) &= a_h(h_d[k]) + a'_{hd}[k] \Delta h_d[k], \\ a'_{hd} &:= \left. \frac{da_h}{dh} \right|_{h_d}, \quad \delta a'_{hd} := \left. \frac{d^2 a_h}{dh^2} \right|_{h_d}, & a'''_{hd} &:= \left. \frac{d^3 a_h}{dh^3} \right|_{h_d}. \end{aligned}$$

Gain-error back-step:

$$\begin{aligned} a_h[k-i] - \hat{a}_h[k-i] &\approx e_{a_h}[k] - (h_d[k] - h_d[k-i]) e_{a'_{hd}}[k] \\ a_h[k] f_{dh}[k] - a_h[k] \bar{\bar{f}}_{dh}[k] &\approx e_{a_h}[k] \underbrace{\left\{ f_{dh}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dh}[k-i] \right\}}_{F_{dh}[k]} \\ &\quad - e_{a'_{hd}}[k] \underbrace{\left\{ (1 - \lambda_c) \sum_{i=1}^d (h_d[k] - h_d[k-i]) f_{dh}[k-i] \right\}}_{F_{dh}^{\Delta H}[k]} \\ &\quad + \delta h_D^d[k] + (1 - \lambda_c) n_h[k] \end{aligned}$$

$$\begin{aligned} \delta h[k+1] &= \lambda_c \cdot \delta h[k] - F_{dh}[k] \cdot e_{a_h}[k] + F_{dh}^{\Delta H}[k] \cdot e_{a'_{hd}}[k] - \delta h_D^d[k] \\ &\quad - \underbrace{\left\{ \delta h_T^d[k] + (1 - \lambda_c) n_h[k] \right\}}_{\varepsilon_h[k]} \end{aligned}$$

True gain recursion

(exact curve values swept to $h_d[k+1]$; $[\cdot]$ = residual the estimator drops)

$$a_h[k+1] = a_h(h_d[k+1]) - a'_{hd}[k+1] \delta h[k+1], \quad a'_{hd}[k+1] = a'_h(h_d[k+1]), \quad \delta a'_{hd}[k+1] = a''_h(h_d[k+1])$$

$$\begin{aligned}
a_h[k+1] &= a_h + a'_{hd} \Delta h_d + (1-\lambda_c) a'_{hd} \delta h + a'_{hd} F_{dh} e_{a_h} + a'_{hd} \varepsilon_h \left[+ \frac{1}{2} \delta a'_{hd} \delta h[k+1]^2 \right] \\
a'_{hd}[k+1] &= a'_{hd} + \delta a'_{hd} \Delta h_d \left[+ \frac{1}{2} a'''_{hd} \Delta h_d^2 \right] \\
\delta a'_{hd}[k+1] &= \delta a'_{hd} + a'''_{hd} \Delta h_d
\end{aligned}$$

True state equation

($\delta h_D^d \equiv 0$; leading order)

$$\begin{aligned}
\delta h_1[k+1] &= \delta h_2[k] \\
\delta h_2[k+1] &= \delta h_3[k] \\
\delta h_3[k+1] &= \lambda_c \delta h_3[k] - F_{dh}[k] e_{a_h}[k] + F_{dh}^{\Delta H}[k] e_{a'_{hd}}[k] - \varepsilon_h[k] \\
a_h[k+1] &= a_h[k] + a'_{hd} \Delta h_d + (1-\lambda_c) a'_{hd} \delta h_3 + a'_{hd} F_{dh} e_{a_h} + a'_{hd} \varepsilon_h \\
a'_{hd}[k+1] &= a'_{hd}[k] + \delta a'_{hd}[k] \Delta h_d \\
\delta a'_{hd}[k+1] &= \delta a'_{hd}[k] + a'''_{hd}[k] \Delta h_d
\end{aligned}$$

Output equation

$$\begin{cases} y_1[k] = \delta h_m[k] = \delta h_1[k] + \underbrace{n_h[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_h[k-d] + n_a[k]}_{a_{hm}[k]} \approx a_h[k] - \Delta H_d[k] a'_{hd}[k] + \underbrace{n_a[k]}_{r_2[k]}, \quad \Delta H_d[k] := h_d[k] - h_d[k-d] \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\Delta H_d[k] & 0 \end{bmatrix}, \quad \mathbf{y}[k] = \mathbf{H} \mathbf{x}[k] + \mathbf{r}[k]$$

(the thermal-based a_{hm} additionally carries a curvature bias $\frac{1}{2} \delta a'_{hd} \sigma_h^2$ from the thermal excursion σ_h^2 ; folded into r_2/R_2 , deferred.)

Estimator (model)

(truncated: $\hat{a}_h$ 1st-order, $\hat{a}'_{hd}$ swept, $\delta \hat{a}'_{hd}$ frozen $\Rightarrow$ drops the true $a'''_{hd} \Delta h_d$)

$$e_{\delta h_1} = \delta h_1 - \hat{\delta h}_1, \quad e_{a_h} = a_h - \hat{a}_h, \quad e_{a'_{hd}} = a'_{hd} - \hat{a}'_{hd}, \quad e_{\delta a'_{hd}} = \delta a'_{hd} - \delta \hat{a}'_{hd}$$

$$e_{y_1} = y_1 - \hat{y}_1 = \delta h_m - \delta \hat{h}_1 = e_{\delta h_1} + \underbrace{n_h}_{r_1}$$

$$e_{y_2} = y_2 - \hat{y}_2 = a_{hm} - (\hat{a}_h - \Delta H_d \hat{a}'_{hd}) = e_{a_h} - \Delta H_d e_{a'_{hd}} + \underbrace{n_a}_{r_2}$$

$$\delta \hat{h}_1[k+1] = \delta \hat{h}_2 + \ell_{11} e_{y_1} + \ell_{12} e_{y_2}$$

$$\delta \hat{h}_2[k+1] = \delta \hat{h}_3 + \ell_{21} e_{y_1} + \ell_{22} e_{y_2}$$

$$\delta \hat{h}_3[k+1] = \lambda_c \delta \hat{h}_3 + \ell_{31} e_{y_1} + \ell_{32} e_{y_2}$$

$$\hat{a}_h[k+1] = \hat{a}_h + \hat{a}'_{hd} \Delta h_d + (1-\lambda_c) \hat{a}'_{hd} \delta \hat{h}_3 + \ell_{41} e_{y_1} + \ell_{42} e_{y_2}$$

$$\hat{a}'_{hd}[k+1] = \hat{a}'_{hd} + \delta \hat{a}'_{hd} \Delta h_d + \ell_{51} e_{y_1} + \ell_{52} e_{y_2}$$

$$\delta \hat{a}'_{hd}[k+1] = \delta \hat{a}'_{hd} + \ell_{61} e_{y_1} + \ell_{62} e_{y_2}$$

Error dynamics

(True – Estimator)

$$e_{\delta h_1}[k+1] = e_{\delta h_2} - \ell_{11} e_{\delta h_1} - \ell_{12} e_{a_h} + \ell_{12} \Delta H_d e_{a'_{hd}}$$

$$e_{\delta h_2}[k+1] = e_{\delta h_3} - \ell_{21} e_{\delta h_1} - \ell_{22} e_{a_h} + \ell_{22} \Delta H_d e_{a'_{hd}}$$

$$e_{\delta h_3}[k+1] = \lambda_c e_{\delta h_3} - F_{dh} e_{a_h} + F_{dh}^{\Delta H} e_{a'_{hd}} - \underbrace{\varepsilon_h}_{q_3}$$

$$- \ell_{31} e_{\delta h_1} - \ell_{32} e_{a_h} + \ell_{32} \Delta H_d e_{a'_{hd}}$$

$$e_{a_h}[k+1] = (1+a'_{hd}F_{dh}) e_{a_h} + (1-\lambda_c)a'_{hd} e_{\delta h_3} + \Delta h_d e_{a'_{hd}} + \underbrace{a'_{hd} \varepsilon_h}_{q_4}$$

$$- \ell_{41} e_{\delta h_1} - \ell_{42} e_{a_h} + \ell_{42} \Delta H_d e_{a'_{hd}}$$

$$e_{a'_{hd}}[k+1] = e_{a'_{hd}} + \Delta h_d e_{\delta a'_{hd}} - \ell_{51} e_{\delta h_1} - \ell_{52} e_{a_h} + \ell_{52} \Delta H_d e_{a'_{hd}}$$

$$e_{\delta a'_{hd}}[k+1] = e_{\delta a'_{hd}} + \underbrace{a'''_{hd} \Delta h_d}_{q_6} - \ell_{61} e_{\delta h_1} - \ell_{62} e_{a_h} + \ell_{62} \Delta H_d e_{a'_{hd}}$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh} & F_{dh}^{\Delta H} & 0 \\ 0 & 0 & (1-\lambda_c)a'_{hd} & 1+a'_{hd}F_{dh} & \Delta h_d & 0 \\ 0 & 0 & 0 & 0 & 1 & \Delta h_d \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k]$$

$\mathbf{q} = [0, 0, q_3, q_4, 0, q_6]^\top$: $q_3 = -\varepsilon_h$, $q_4 = a'_{hd} \varepsilon_h$ (thermal); $q_6 = a'''_{hd} \Delta h_d$ (the true 3rd-order sweep the frozen $\delta \hat{a}'_{hd}$ misses). $Q_{33} = \sigma_\varepsilon^2$, $Q_{44} = a'_{hd}{}^2 \sigma_\varepsilon^2$, $Q_{55} = 0$.

Closure for Q_{66} : taking $Q_{66} = a'''_{hd}{}^2 \Delta h_d^2$ needs a'''_{hd} , whose own noise needs $a^{(4)}, a^{(5)}, \dots$ — an unclosed derivative regress. Since $a^{(n)} = (a_{nom}/R^n) D^{(n)}$, each derivative scales by $1/R$, so close the hierarchy at the top state via $a'''_{hd} \sim \delta a'_{hd}/R$: $\boxed{Q_{66} \approx (\delta a'_{hd} \Delta h_d / R)^2}$ — uses only the estimated $\delta \hat{a}'_{hd}$ and known $R, \Delta h_d$ (no a'''_{hd}); trajectory-gated ($\propto \Delta h_d^2$, zero on a hold) and self-scaling to the curvature.

